INCH-POUND
MIL-DTL-24643/23H
27 November 2023
SUPERSEDING
MIL-DTL-24643/23G
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 300 VOLTS, TYPE LSTTSU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First - Copper conductor, coated or uncoated, 22 AWG, class K stranding.

Second - Thermoset insulation, minimum average wall thickness of 0.017 inch. Insulation shall be colored (see fourth) if insulation covering is transparent. Insulation shall be colored white if insulation covering is colored.

Third - Insulation covering of polyamide, in accordance with ASTM D4066, colored (see fourth) or transparent at manufacturer's option. Minimum average wall thickness of 0.003 inch. Diameter over insulated and covered conductor shall be 0.076 inch minimum. Twisted pair identification code applied by method 6.

Fourth - Two conductors, one white, one black, twisted together to form a pair having a left-hand lay length not greater than 2¼ inches. For type LSTTSU: 1½ inches only, three conductors (one white, one black, and one red) twisted together to form a triad having a lay length no greater than 2¼ inches.

Fifth - The specified number of pairs (see table I) shall be cabled together with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Outermost layer shall be left-hand. Fillers may be used as necessary to form a firm, well-rounded assembly.

Sixth - An optional binder.

Seventh - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/23	Type and size	Number of pairs	Jacket thickness min. avg. (inch)	Cable overall diameter max. (inches)
-01UN	LSTTSU-1½	1½	0.050	0.330
-02UN	LSTTSU-3	3	0.050	0.450
-03UN	LSTTSU-5	5	0.050	0.540
-04UN	LSTTSU-10	10	0.062	0.675
-05UN	LSTTSU-15	15	0.062	0.800
-06UN	LSTTSU-20	20	0.062	0.870
-07UN	LSTTSU-30	30	0.075	1.080
-08UN	LSTTSU-40	40	0.075	1.200
-09UN	LSTTSU-50	50	0.075	1.400
-10UN	LSTTSU-60	60	0.075	1.450

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	18.60
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Abrasion resistance (insulation scrapes, min.)	250
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50

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Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Gas flame (circuit integrity) (1 hour) (see note below)	No failure

NOTE: For gas flame test of LSTTSU-1½, the three connectors shall be connected to phases A, B, and C of the power supply. For tests of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

a. Pair #1: One conductor to phase A, one conductor to phase B.
b. Pair #2: One conductor to phase C, one conductor to phase A.
c. Pair #3: One conductor to phase B, one conductor to phase C.
d. Pair #4: One conductor to phase A, one conductor to phase B.
e. Pair #5: One conductor to phase C, one conductor to phase A.
f. Pair #6 through pair #60: In similar sequence.

Flame propagation (cable)	No failure

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Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours) (see note)	
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTTSU	1½ through 10	2,500
LSTTSU	15 and 20	1,500
LSTTSU	30 through 60	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-039)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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